

SIH 26143 — 36 Detailed Work Packages

6 phases × 6 people · detailed responsibilities, deliverables, dependencies and acceptance conditions

P1 — Remote Sensing + AI/ML

Own the SAR/ML detection layer and make model outputs measurable, georeferenceable and usable by reconstruction and attribution.

Phase	Detailed work package
PHASE 1 — DATA & PREPROCESSING	Sentinel-1 preprocessing pipeline Set up the SAR ingestion structure; inspect VV/VH channels; perform radiometric preparation, normalization and appropriate speckle handling; preserve scene metadata and create reproducible preprocessing configurations; generate model-ready tiles without breaking image/mask correspondence. Deliverable: reproducible preprocessed SAR inputs + preprocessing configuration + sample before/after checks. Acceptance: same input/config produces the same model-ready output and no image/mask pairing is lost.
PHASE 2 — SPILL DETECTION	U-Net segmentation model Build the baseline U-Net in PyTorch; define input channels, loss function, train/validation loop, augmentation policy and checkpointing; train on the primary binary-mask dataset; save inference weights and configuration. Deliverable: trained U-Net checkpoint + inference script + training configuration. Acceptance: inference produces a binary/soft spill mask for an unseen scene.
PHASE 3 — SOURCE RECONSTRUCTION	Spill morphology features Convert segmentation output into measurable geometry: connected components, area, centroid, bounding geometry, orientation and shape descriptors; distinguish candidate regions and retain confidence information for downstream processing. Deliverable: structured spill-feature object containing geometry, area and model confidence. Acceptance: dashboard/backend can consume the same geometry and metadata.
PHASE 4 — AIS CORRELATION	Detection-to-AIS feature interface Define the exact hand-off from detection/reconstruction to AIS analysis: incident ID, detection time, spill geometry, reconstructed source corridor, origin-time window and uncertainty fields; document units and coordinate systems. Deliverable: versioned feature contract/schema for the attribution layer. Acceptance: P4 can run correlation without manually reformatting ML output.
PHASE 5 — EVIDENCE FUSION + UI	Detection confidence integration Expose segmentation confidence and look-alike indicators as evidence inputs; separate model confidence from physical/source confidence; ensure low-quality or ambiguous detections can be flagged rather than silently treated as confirmed oil. Deliverable: detection-evidence payload + uncertainty/quality flags. Acceptance: ambiguous detection can propagate an explicit warning downstream.
PHASE 6 — VALIDATION & DEPLOYMENT	Final ML evaluation Run held-out evaluation; calculate IoU, Dice, precision and recall; analyse false positives on look-alikes; record failure examples by scene condition; compare against the agreed baseline only after evaluation is complete. Deliverable: ML evaluation report, error gallery and final model card. Acceptance: every reported metric has a defined split and reproducible evaluation script.

P2 — Backend + Database

Own the system backbone: persistence, APIs, orchestration, provenance, security and reliable service-to-service interfaces.

Phase	Detailed work package
PHASE 1 — DATA & PREPROCESSING	Data storage architecture Design PostgreSQL/PostGIS entities for scenes, detections, spill geometries, incidents, source corridors, AIS observations, candidate vessels, evidence features and provenance; define primary/foreign keys, spatial indexes and timestamps. Deliverable: schema/ERD + migrations + indexing plan. Acceptance: all core outputs can be persisted with provenance.
PHASE 2 — SPILL DETECTION	Detection API + persistence Build the API endpoint/workflow for SAR scene submission and detection retrieval; validate inputs; store mask metadata, geometry, confidence and processing status; expose incident IDs for downstream services. Deliverable: working detection API + persistence layer + API examples. Acceptance: a scene can move from request to stored detection result.
PHASE 3 — SOURCE RECONSTRUCTION	Reconstruction service Implement the service contract between detection, environmental forcing and drift modelling; create job/status handling, result persistence and failure states; store source corridor, origin window and uncertainty metadata. Deliverable: reconstruction API/service workflow. Acceptance: successful and failed reconstruction jobs are both represented explicitly.
PHASE 4 — AIS CORRELATION	AIS data service Create AIS ingestion/query endpoints; validate timestamps and coordinates; index vessel observations by time and geography; support queries around a reconstructed source corridor and origin window. Deliverable: AIS storage + query API + indexed trajectory retrieval. Acceptance: API returns only observations matching requested spatial/temporal bounds.
PHASE 5 — EVIDENCE FUSION + UI	Evidence-fusion backend Create models for evidence families, candidate vessels, feature values, evidence provenance, uncertainty, ranking state and abstention; implement deterministic scoring/fusion configuration rather than hiding logic in UI code. Deliverable: evidence-fusion service + auditable stored results. Acceptance: candidate result can be traced back to component evidence.
PHASE 6 — VALIDATION & DEPLOYMENT	Backend integration + security Connect all service endpoints; add authentication/authorization, input validation, error handling, logging, database constraints and audit events; test invalid inputs and partial service failures. Deliverable: integrated backend + security checklist + API test suite. Acceptance: core workflow runs end-to-end and failures return explicit states instead of fabricated outputs.

P3 — Frontend + Product + Geospatial UI

Own the investigator-facing geospatial product and make every important evidence layer understandable on one screen.

Phase	Detailed work package
PHASE 1 — DATA & PREPROCESSING	Data-to-UI specification Map every backend field to a visible dashboard element: scene metadata, spill mask, confidence, source corridor, origin window, drift particles, vessel track, evidence family and uncertainty; define empty/error states. Deliverable: UI data contract + screen/state specification. Acceptance: no UI field is invented or disconnected from a backend source.
PHASE 2 — SPILL DETECTION	Detection visualisation Build SAR scene view with spill-mask overlay, legend, confidence display and incident metadata; add controls for original imagery versus model output; make clear which layers are observed and which are model-generated. Deliverable: functional detection map component. Acceptance: user can distinguish SAR observation from predicted mask.
PHASE 3 — SOURCE RECONSTRUCTION	Source/drift visualisation Display detected slick, backward particles, source corridor, origin-time window and forward drift corridor; add uncertainty styling and explanatory labels; avoid presenting the reconstructed source as an exact point. Deliverable: reconstruction map component. Acceptance: uncertainty and time-window information remain visible at all relevant zoom levels.
PHASE 4 — AIS CORRELATION	Vessel-track UI Render historical AIS tracks and positions around the reconstructed source region; show vessel metadata, time filtering, AIS-gap indicators and track continuity; support toggling evidence layers. Deliverable: AIS trajectory map layer + controls. Acceptance: candidate tracks are synchronized to the incident time window.
PHASE 5 — EVIDENCE FUSION + UI	Investigator dashboard Assemble the full investigation view: SAR, spill polygon, source corridor, drift, AIS tracks, candidate list, evidence cards and timeline; label synthetic/demo content explicitly when applicable. Deliverable: integrated investigator dashboard. Acceptance: one incident can be followed from detection to candidate evidence without switching tools.
PHASE 6 — VALIDATION & DEPLOYMENT	End-to-end UI integration Replace static placeholders where real outputs exist; add loading, error, insufficient-evidence and no-data states; verify every number shown on the screen is sourced from the backend or clearly labelled illustrative. Deliverable: presentation-ready working UI connected to the actual pipeline. Acceptance: demo scenario runs without hidden/manual data edits.

P4 — GIS + AIS + Attribution

Own AIS trajectory reasoning, GIS correlation and the explainable candidate-prioritisation features.

Phase	Detailed work package
PHASE 1 — DATA & PREPROCESSING	AIS data schema + cleaning Define and validate MMSI/IMO, timestamp, latitude, longitude, SOG, COG, heading and vessel metadata; remove duplicates/out-of-range points; preserve source timestamps and provenance; document region/source limitations. Deliverable: cleaned AIS schema + validation rules. Acceptance: invalid observations are flagged/rejected and provenance is retained.
PHASE 2 — SPILL DETECTION	Geospatial detection handling Transform segmentation geometry into geographic coordinates; validate coordinate reference systems; create spatial footprints for spill and source regions; prepare spatial queries for vessel intersections. Deliverable: georeferenced spill/source geometries and spatial query functions. Acceptance: geometry overlays correctly on the map and spatial queries return expected test cases.
PHASE 3 — SOURCE RECONSTRUCTION	GIS spatial preparation Create spatial indexes, buffers/corridors and temporal-spatial query helpers; prepare basemap/reference layers; define distance calculations and coordinate conventions for evidence features. Deliverable: reusable GIS utility layer. Acceptance: distance/intersection calculations pass controlled test cases.
PHASE 4 — AIS CORRELATION	Trajectory reconstruction + filtering Order observations by vessel/time; detect gaps and discontinuities; interpolate only when justified; calculate distance to source corridor, time difference, heading/track compatibility and continuity; preserve AIS-quality flags. Deliverable: reconstructed vessel trajectories + evidence features. Acceptance: every candidate has traceable spatial, temporal and trajectory values.
PHASE 5 — EVIDENCE FUSION + UI	Explainable candidate features Implement spatial proximity, temporal overlap, trajectory compatibility and AIS reception-quality features; define normalization and missing-data behaviour; provide feature-level explanations instead of a single opaque score. Deliverable: candidate feature vector + explanation payload. Acceptance: a reviewer can see why each feature increased/decreased investigative relevance.
PHASE 6 — VALIDATION & DEPLOYMENT	Attribution scenario testing Create controlled synthetic vessel/source scenarios with known ground truth; run the candidate prioritisation process; calculate Top-1, Top-3 and MRR where appropriate; document cases where AIS gaps or ambiguity prevent a reliable ranking. Deliverable: attribution evaluation dataset + results + failure analysis. Acceptance: ground truth is known by construction and results are reproducible.

P5 — Environmental + Integration + DevOps

Own environmental forcing, Lagrangian source reconstruction, uncertainty propagation and reproducible deployment.

Phase	Detailed work package
PHASE 1 — DATA & PREPROCESSING	Metoccean ingestion Acquire and standardise ERA5 wind and CMEMS current data for selected study windows; align timestamps, spatial grids and units; cache forcing fields; record source/version metadata. Deliverable: standardised wind/current forcing inputs. Acceptance: forcing can be queried consistently for any test timestamp.
PHASE 2 — SPILL DETECTION	Environmental context layer Attach wind/current context to detected incidents; record forcing availability and missing-data conditions; flag environmental regimes that may increase interpretation uncertainty. Deliverable: environmental-context object + quality flags. Acceptance: each reconstruction input has a documented forcing source/time.
PHASE 3 — SOURCE RECONSTRUCTION	OpenDrift/OpenOil modelling Configure Lagrangian particle simulations using wind/current forcing; implement backward hindcast from detected slick and forward drift forecast; define release/particle settings; capture simulation uncertainty and forcing gaps. Deliverable: reproducible drift-model workflow producing particle trajectories/corridors. Acceptance: same incident/forcing configuration reproduces the stored trajectory result.
PHASE 4 — AIS CORRELATION	Environmental trajectory context Convert particle output into investigator-friendly source/drift corridors, origin-time candidates and uncertainty bounds; pass these to P4 without converting model output into a claim of exact causation. Deliverable: reconstruction result schema for AIS correlation. Acceptance: P4 receives a corridor/time window plus uncertainty, not an unjustified exact source point.
PHASE 5 — EVIDENCE FUSION + UI	Uncertainty + abstention integration Define what happens when forcing is missing, the source corridor becomes too broad, AIS coverage is inadequate or evidence families disagree; propagate uncertainty and implement explicit insufficient-evidence states. Deliverable: uncertainty/abstention rules + integrated status signals. Acceptance: insufficient evidence produces abstention rather than a forced candidate.
PHASE 6 — VALIDATION & DEPLOYMENT	Deployment + system testing Containerise services with Docker Compose; configure environment variables and reproducible startup; test service failures, timeouts and resource limits; measure practical prototype latency and document scope. Deliverable: deployable prototype stack + system test report. Acceptance: clean environment can start the system and execute the controlled demo workflow.

P6 — Validation + QA + Scientific Evaluation

Own quantitative validation and scientific QA so every major claim has a defined test and every failure is visible.

Phase	Detailed work package
PHASE 1 — DATA & PREPROCESSING	Dataset audit Audit image/mask correspondence, class/label structure, file integrity, metadata, train/validation/test separation and possible leakage; document what the dataset actually contains and what it does not contain. Deliverable: dataset audit sheet + approved split manifest. Acceptance: split is fixed before final metric reporting.
PHASE 2 — SPILL DETECTION	Detection evaluation Build the evaluation script for IoU, Dice, precision, recall and false-positive analysis; evaluate held-out data; create representative error cases and distinguish segmentation errors from look-alike confusion. Deliverable: reproducible detection metrics + error gallery. Acceptance: metrics can be regenerated from the saved checkpoint and split.
PHASE 3 — SOURCE RECONSTRUCTION	Source-recovery validation Generate controlled synthetic release/source scenarios; run backward reconstruction; measure corridor/time-window containment of the known source; test sensitivity to forcing uncertainty and observation perturbations. Deliverable: source-recovery test set + containment/error report. Acceptance: ground-truth source and release time are known by construction.
PHASE 4 — AIS CORRELATION	AIS-gap testing Create controlled AIS deletion scenarios; remove observations at different gap patterns; rerun correlation and measure changes in candidate set/ranking; verify missing data does not receive adverse evidence weight. Deliverable: AIS robustness report. Acceptance: behaviour under missing AIS is quantified and documented.
PHASE 5 — EVIDENCE FUSION + UI	Fusion evaluation Compare evidence-family behaviour individually and in combination; test ranking stability, sensitivity to missing evidence and abstention thresholds; ensure results are explainable and reproducible. Deliverable: evidence-fusion evaluation matrix. Acceptance: every fusion test has a defined input scenario and expected behaviour.
PHASE 6 — VALIDATION & DEPLOYMENT	Final validation + QA report Run the complete matrix across detection, look-alikes, source recovery, AIS gaps, attribution scenarios, abstention, reproducibility and failure handling; freeze reported metrics; document limitations and unresolved risks. Deliverable: final QA report + validation checklist + demo sign-off. Acceptance: every major claim in the PPT maps to a test, measured result or explicit planned item.